

AI Engineer Bootcamp - Sample Report

Large Language Models (LLMs) have transformed the way we interact with AI systems. Retrieval-Augmented Generation (RAG) combines the power of LLMs with external knowledge retrieval to produce more accurate, up-to-date, and verifiable responses.

This sample document is used to demonstrate PDF processing techniques including text extraction, table detection, and chunk generation for vector databases.

Model Performance Comparison

Model	Params	MMLU	HumanEval	Cost/1M
GPT-4o	1.8T	88.7	90.2	\$5.00
Claude Opus 4.7	~2T	86.8	84.9	\$15.00
DeepSeek-V3	671B	88.5	82.6	\$0.27
Gemini 2.5 Pro	~1T	90.0	88.1	\$7.00
Llama 3.3 70B	70B	86.0	80.5	\$0.59

RAG Pipeline Overview

A typical RAG pipeline consists of the following stages:

- 1. Document Ingestion - Load PDFs, Word docs, web pages, and other sources.
- 2. Text Extraction - Parse raw text from each document type.
- 3. Chunking - Split text into overlapping segments (typically 256-512 tokens).
- 4. Embedding - Convert each chunk into a dense vector representation.
- 5. Indexing - Store vectors in a database like Chroma, FAISS, or Pinecone.
- 6. Retrieval - At query time, embed the question and find top-k similar chunks.
- 7. Generation - Feed retrieved context + query to the LLM for a final answer.

Chunk size and overlap are critical hyperparameters. Too small loses context; too large dilutes relevance. A 400-token chunk with 50-token overlap works well for most knowledge-base use cases.

Chunking Strategy Comparison

Strategy	Pros	Cons
Fixed Token	Simple, predictable	May break sentences
Sentence	Preserves meaning	Variable chunk size
Semantic	Best coherence	Slower, needs embedder
Recursive	Respects structure	Complex to tune

Embedding Models & Vector Search

Embedding models convert text into high-dimensional vectors that capture semantic meaning. Similar texts produce similar vectors, enabling semantic search.

Popular embedding models include OpenAI text-embedding-3-large (3072 dims), Snowflake Arctic Embed L v2 (1024 dims), and BGE-M3 for multilingual tasks.

Vector similarity is measured using cosine similarity or dot product. Approximate Nearest Neighbor (ANN) algorithms like HNSW allow sub-millisecond retrieval even over millions of vectors.